

# The Backtest Fallacy: Why Copyable Returns Are Already Consumed

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*A Throughput-Based Analysis Under the Theory of Necessary Action (TNA)*

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## Abstract

Backtesting is widely used to validate trading strategies and to market copy-trading systems. However, backtests implicitly assume infinite liquidity, negligible market impact, and observer-independent execution.

This paper demonstrates that these assumptions violate fundamental throughput constraints in real markets.

Using the Theory of Necessary Action (TNA), we show that any strategy whose returns are observable, replicable, and scalable necessarily exhausts the market's capacity to absorb synchronized orders.

We formalize this limitation as a saturation of negentropy throughput ( $\Psi$ ), leading to an accumulation of incoherence ( $\aleph$ ) and inevitable performance collapse upon replication.

The central result is concise: **any return that can be copied at scale has already been consumed.**

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## 1. Introduction

Backtests are backward-looking measurements of performance under historical flow conditions. They do not model the causal effect of observation, replication, or synchronization.

Copy-trading platforms implicitly assert:

*If a strategy worked in the past, it can be followed in the future.*

This paper shows why this assertion is physically false.

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## 2. The Hidden Assumptions of Backtesting

Standard backtests assume:

1. Liquidity is sufficient to absorb arbitrarily scaled orders
2. Execution does not alter prices
3. Other agents do not respond to identical signals
4. Strategy performance is invariant under replication

These assumptions define an **observer-free market**, which does not exist.

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## 3. Throughput Constraints in Real Markets

In TNA terms:

- $\aleph$  (Aleph): accumulated incoherence (latent market stress, valuation gaps, expectation load)
- $\Psi$  (Psi): market throughput — the capacity to convert nominal value into liquidity without destabilizing price

Markets operate under **finite  $\Psi$** .

Order flow consumes  $\Psi$ .

Synchronized flow consumes it nonlinearly.

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## 4. The Copying Paradox

Let a strategy  $S$  produce positive returns under historical flow  $F_0$ .

When  $S$  is observed and replicated:

- Total flow becomes  $F = k \cdot F_0$
- Market impact grows faster than  $k$
- $\Psi$  is saturated before execution completes

Thus:

- Early followers profit

- Late followers pay the price impact
- Aggregate returns converge to zero or negative

This is not behavioral.

It is **throughput physics**.

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## 5. Backtests as Oniric Valuations

Backtest results measure:

*performance under a flow regime that no longer exists*

They ignore:

- order book depth
- slippage under scale
- reflexive demand
- feedback between observation and execution

Thus, backtests generate **oniric returns**:

nominal profitability without realizable liquidity.

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## 6. Formal Proof of the Backtest Saturation Theorem

*“The following theorem is not a behavioral claim but a consequence of finite market throughput under replication.”*

### Definitions

Let:

- $\Psi$  = market throughput, defined as the maximum rate at which nominal value can be converted into liquidity without destabilizing prices.
- $\mathcal{F}$  = order flow generated by a trading strategy.
- $\mathcal{F}_0$  = historical order flow under which the backtest was measured.

- $k \in \mathbb{N}$  = number of agents replicating the strategy.
- $\mathcal{F}(k) = k \cdot \mathcal{F}_0$  = replicated flow.
- $\aleph$  = accumulated incoherence, defined as unrealizable nominal value caused by execution friction, slippage, and price impact.

Assume markets satisfy the empirical constraint:

$$\frac{\partial \Psi}{\partial \mathcal{F}} < 0 \quad \text{for sufficiently large } \mathcal{F}$$

That is, throughput is finite and degrades under excessive synchronized flow.

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## Lemma 1: Nonlinearity of Market Impact

Market impact  $I(\mathcal{F})$  is a strictly convex function of order flow magnitude:

$$I(\mathcal{F}) \sim \mathcal{F}^\alpha \quad \text{with } \alpha > 1$$

This is empirically established across equity, FX, and futures markets.

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## Lemma 2: Throughput Saturation

There exists a finite  $\Psi_{\max}$  such that for any replicated flow  $\mathcal{F}(k)$ :

$$\exists k^* : \mathcal{F}(k^*) \geq \Psi_{\max}$$

Beyond this point, execution requires price displacement rather than liquidity absorption.

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## Theorem (Backtest Saturation)

**Any trading strategy whose returns are observable and replicable at scale necessarily collapses in performance when replication exceeds market throughput capacity.**

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## Proof

1. A backtest measures returns under historical flow  $\mathcal{F}_0$ , implicitly assuming:

$$\mathcal{F}_0 \ll \Psi_{\max}$$

2. When the strategy is replicated by  $k$  agents, total flow becomes:

$$\mathcal{F}(k) = k \cdot \mathcal{F}_0$$

3. By Lemma 2, there exists  $k^*$  such that:

$$\mathcal{F}(k^*) \geq \Psi_{\max}$$

4. For  $k \geq k^*$ , execution cannot occur without price impact:

$$I(\mathcal{F}(k)) \uparrow \Rightarrow \text{realized returns} \downarrow$$

5. The excess nominal value implied by the backtest becomes unrealizable, accumulating as incoherence:

$$\frac{d\mathfrak{N}}{dk} > 0$$

6. Aggregate returns converge to zero or negative as price impact dominates.

Therefore, **replication destroys the very conditions that produced the backtested returns.**

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## Corollary 1: Inverse Relation Between Alpha and Scalability

Let  $A$  denote excess return (alpha). Then:

$$\frac{\partial A}{\partial k} < 0 \quad \text{for } k \geq k^*$$

High alpha implies low scalability; scalable strategies cannot maintain alpha.

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## Corollary 2: Backtests Measure Exhausted Throughput

Backtests do not measure future opportunity, but **historical throughput already consumed.**

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## Interpretation (Non-Behavioral)

This result does **not** depend on psychology, reflexivity, or expectations.

It follows directly from:

- finite liquidity
- convex market impact
- synchronized execution

The failure of copyable strategies is therefore **physical**, not informational.

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## 7. Empirical Signatures of Impending Collapse

Observable indicators include:

- declining performance as follower count increases
- rising slippage per trade
- shrinking order book depth at execution levels
- increasing correlation between trades
- widening bid–ask spreads under identical signals

These precede drawdowns systematically.

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## 8. Implications

For investors

Past performance is not invalid — **it is already spent.**

For platforms

Publishing strategies destroys their future return capacity.

For regulators

Systemic risk increases with synchronization, not leverage alone.

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## 9. Conclusion

Backtesting does not fail statistically.

It fails **physically**.

Markets are not informational systems; they are throughput-limited systems.

Returns do not disappear because strategies are wrong — they disappear because they are followed.

**The rentability that can be copied has already been consumed.**